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Inventor(s)	Takayama; Hidetoshi et al.

Cell culture system and cell culture method

Abstract

A cell culture system includes: a plurality of cell culture channels each of which includes a fluidic device in which cells are cultured, a pump which makes a liquid flow into the fluidic device, a detection unit which detects a state of the cells that are cultured, and a controller which controls the pump and the detection unit; a measurement unit that measures culture environment of the cells which are cultured in the plurality of cell culture channels; and an information processing apparatus that adjusts the culture environment such that an absolute value of a difference between evaluation values for the states of the cells that are cultured in the plurality of cell culture channels is a threshold value or less, on the basis of the state of the cells detected by the detection unit and the culture environment of the cells measured by the measurement unit.

Inventors: Takayama; Hidetoshi (Kanagawa, JP), Ito; Koju (Kanagawa, JP), Oba; Takahiro (Kanagawa, JP), Oku; Keisuke (Kanagawa, JP)

Applicant: FUJIFILM CORPORATION (Tokyo, JP)

Family ID: 72520883

Assignee: FUJIFILM CORPORATION (Tokyo, JP)

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Primary Examiner: Whatley; Benjamin R

Assistant Examiner: Thompson; Curtis A

Attorney, Agent or Firm: SOLARIS Intellectual Property Group, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/JP2020/007992, filed Feb. 27, 2020, the disclosure of which is incorporated herein by reference in its entirety. Further, this application claims priority from Japanese Patent Application No. 2019-051925, filed Mar. 19, 2019, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a cell culture system and a cell culture method.

RELATED ART

(2) As a technique related to cell culture, a culture station that is used to culture biological cells in a microfluidic device has been known (see JP2018-512853A). The culture station is provided with a pump that makes a culture medium from a culture medium source flow into the microfluidic device and a thermal regulation system that controls the temperature of the microfluidic device.

(3) Incidentally, in the cell culture using a fluidic device, such as a microfluidic device, various functions can be customized through examination of various flow passage structures of the fluidic device. As a result, the fluidic device has been applied to applications such as drug discovery, toxicity evaluation, an organ-on-a-chip, a body-on-a-chip, and analytical chemistry. However, since the fluidic device alone cannot be used for the intended application, a system capable of controlling a liquid feeding system, sensing, adjustment of culture environment, and the like has been required.

(4) Further, in the above-mentioned applications, it is preferable that cell culture using a plurality of fluidic devices can be performed. Furthermore, in this case, it is preferable that the culture

environment of each of the plurality of fluidic devices can be adjusted in an integrated manner while referring to the culture state of the cells that are cultured in each of the plurality of fluidic devices. However, the technique described in JP2018-512853A has not considered that the culture environment of each of the plurality of fluidic devices is adjusted in an integrated manner.

SUMMARY

(5) The present disclosure has been made in view of the above circumstances, and an object thereof is to provide a cell culture system and a cell culture method in which the culture environment of each of a plurality of fluidic devices can be adjusted in an integrated manner.

(6) In order to achieve the above-mentioned object, a cell culture system according to the present disclosure includes: a plurality of cell culture channels each of which includes a fluidic device in which cells are cultured, a liquid feeding unit which makes a liquid flow into the fluidic device, a detection unit which detects a state of the cells that are cultured in the fluidic device, and a controller which controls the liquid feeding unit and the detection unit; a measurement unit that measures culture environment of the cells which are cultured in the plurality of cell culture channels; and an integrated controller that adjusts the culture environment such that an absolute value of a difference between evaluation values for the states of the cells that are cultured in the plurality of cell culture channels is a threshold value or less, on the basis of the state of the cells detected by the detection unit and the culture environment of the cells measured by the measurement unit.

(7) Meanwhile, in order to achieve the above-mentioned object, the cell culture method according to the present disclosure using a cell culture system that includes a plurality of cell culture channels each of which includes a fluidic device in which cells are cultured, a liquid feeding unit which makes a liquid flow into the fluidic device, a detection unit which detects a state of the cells that are cultured in the fluidic device, and a controller which controls the liquid feeding unit and the detection unit, the cell culture method includes: measuring culture environment of the cells which are cultured in the plurality of cell culture channels; and adjusting the culture environment such that an absolute value of a difference between evaluation values for the states of the cells that are cultured in the plurality of cell culture channels is a threshold value or less, on the basis of the state of the cells detected by the detection unit and the measured culture environment of the cells.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram showing an example of a configuration of a cell culture system according to an embodiment.

(2) FIG. 2 is a diagram for explaining a disposition state of a cell culture channel according to the embodiment.

(3) FIG. 3 is a block diagram showing an example of a hardware configuration of an information processing apparatus according to the embodiment.

(4) FIG. 4 is a block diagram showing an example of a functional configuration of the information processing apparatus according to the embodiment in a learning phase.

(5) FIG. 5 is a diagram for explaining information representing a disposition position of the cell culture channel according to the embodiment.

(6) FIG. 6 is a diagram for explaining a learned model according to the embodiment.

(7) FIG. 7 is a flowchart showing an example of learning processing according to the embodiment.

(8) FIG. 8 is a block diagram showing an example of a functional configuration of the information processing apparatus according to the embodiment in an operation phase.

(9) FIG. 9 is a flowchart showing an example of cell culture processing according to the embodiment.

DETAILED DESCRIPTION

(10) Hereinafter, an exemplary embodiment of the technique according to the present disclosure will be described in detail with reference to the drawings.

(11) First, the configuration of a cell culture system **10** according to the present embodiment will be described with reference to FIG. **1**. As shown in FIG. **1**, the cell culture system **10** includes an information processing apparatus **12** and an incubator **14**. Examples of the information processing apparatus **12** include a personal computer, a server computer, and the like. The cell culture system **10** is used in applications such as drug discovery, toxicity evaluation, an organ-on-a-chip, a body-on-a-chip, and analytical chemistry.

(12) A plurality of cell culture channels **16**, a controller **40**, an environment measurement unit **42**, and an environment adjustment unit **44** are provided inside the incubator **14**. The controller **40** is connected to the environment measurement unit **42** and the environment adjustment unit **44**. In the present embodiment, an example of an aspect in which the controller **40**, the environment measurement unit **42**, and the environment adjustment unit **44** are provided in a set inside the incubator **14** will be described, but the present disclosure is not limited thereto. The controller **40**, the environment measurement unit **42**, and the environment adjustment unit **44** may be provided, for example, in a set for each one cell culture channel **16** or a set for each two cell culture channels **16**.

(13) The cell culture channel **16** includes a reservoir **20**, a pump **22**, a fluidic device **24**, a waste liquid unit **26**, a flow rate measurement unit **28**, a micro-differential pressure measurement unit **30**, a potential of hydrogen (pH) measurement unit **32**, a resistance measurement unit **34**, an imaging unit, **36** and a controller **38**. The controller **38** is connected to the pump **22**, the flow rate measurement unit **28**, the micro-differential pressure measurement unit **30**, the pH measurement unit **32**, the resistance measurement unit **34**, and the imaging unit **36**.

(14) A liquid is stored in the reservoir **20**. Examples of the liquid stored in the reservoir **20** include a liquid medium, a cell suspension, an added compound solution, an evaluation chemical agent, and a tracer solution. The pump **22** is, for example, a micropump that feeds the liquid by driving a piezo element, and makes the liquid stored in the reservoir **20** flow into the fluidic device **24**. The pump **22** is an example of a liquid feeding unit that makes the liquid flow into the fluidic device **24**. In the pump **22**, the flow rate of the liquid is controlled by the controller **38**. In a case where there are a plurality of types of liquids to be fed, the reservoir **20** and the pump **22** are provided in a set for each type of liquid.

(15) In the fluidic device **24**, cells are cultured. The fluidic device **24** includes an inlet into which the liquid flows, an outlet out of which the liquid flows, and a flow passage which connects the inlet and the outlet to each other. Examples of the fluidic device **24** include a microfluidic device having micro flow passages. In the following, cells cultured in the fluidic device **24** will be referred to as “cultured cells”.

(16) The waste liquid unit **26** discards the liquid flowing out of the fluidic device **24**. The cell culture channel **16** may be provided with a circulation unit that circulates the liquid flowing out of the fluidic device **24** to the reservoir **20**, instead of the waste liquid unit **26**.

(17) The flow rate measurement unit **28** measures the flow rate of the liquid flowing through the fluidic device **24** per unit time and outputs the measurement result to the controller **38**. The micro-differential pressure measurement unit **30** measures the micro-differential pressure between the inlet and the outlet of the fluidic device **24** and outputs the measurement result to the controller **38**. The pH measurement unit **32** measures the pH of the liquid flowing through the fluidic device **24** and outputs the measurement result to the controller **38**. The measurement results of the flow rate measurement unit **28**, the micro-differential pressure measurement unit **30**, and the pH measurement unit **32** are used, for example, to detect the presence or absence of contamination inside the fluidic device **24**.

(18) The resistance measurement unit **34** measures the transepithelial electrical resistance of the

cultured cells and outputs the measurement result to the controller **38**. The transepithelial electrical resistance is used to evaluate the barrier function of cells, and numerical values thereof vary depending on the environment temperature, the state of cultured cells, and the like. The imaging unit **36** captures an image of the cultured cells in accordance with a predetermined frame rate and outputs image data showing the image obtained by the capturing of the image to the controller **38**. The resistance measurement unit **34** and the imaging unit **36** are examples of a detection unit that detects the state of cells. Further, the flow rate measurement unit **28**, the micro-differential pressure measurement unit **30**, the pH measurement unit **32**, the resistance measurement unit **34**, and the imaging unit **36** may be formed of various sensors that detect amounts to be measured.

(19) The controller **38** controls the pump **22**, the flow rate measurement unit **28**, the micro-differential pressure measurement unit **30**, the pH measurement unit **32**, the resistance measurement unit **34**, and the imaging unit **36**. Examples of the controller **38** include a processor such as a programmable logic device (PLD).

(20) The controller **40** controls the environment measurement unit **42** and the environment adjustment unit **44**. Examples of the controller **40** include a processor such as a PLD. The environment measurement unit **42** measures the culture environment of the cells that are cultured in each cell culture channel **16**. The culture environment that is measured by the environment measurement unit **42** includes a temperature of the liquid flowing through the fluidic device **24**, an environment temperature around the fluidic device **24**, an environment humidity around the fluidic device **24**, a carbon dioxide concentration around the fluidic device **24**, a nitrogen concentration around the fluidic device **24**, and an oxygen concentration around the fluidic device **24**. That is, the environment measurement unit **42** includes various sensors capable of measuring the culture environment.

(21) The environment adjustment unit **44** adjusts the culture environment of the cultured cells that are cultured in each cell culture channel **16**. The culture environment that can be adjusted by the environment adjustment unit **44** is the same, for example, as the culture environment that can be measured by the environment measurement unit **42**. The controller **38** and the controller **40** are connected to the information processing apparatus **12** and are controlled in an integrated manner by the information processing apparatus **12**. The environment adjustment unit **44** may adjust the culture environment for each cell culture channel **16**. In this case, the environment measurement unit **42** also measures the culture environment for each cell culture channel **16**.

(22) The cell culture channels **16** are arranged side by side in the incubator **14** as shown in FIG. 2 as an example. FIG. 2 is a front view of the incubator **14**.

(23) Next, the hardware configuration of the information processing apparatus **12** according to the present embodiment will be described with reference to FIG. 3. As shown in FIG. 3, the information processing apparatus **12** includes a central processing unit (CPU) **50**, a memory **51** as a temporary storage area, and a non-volatile storage unit **52**. Further, the information processing apparatus **12** includes a display unit **53**, such as a liquid crystal display, an input unit **54**, such as a keyboard and a mouse, and an external interface (I/F) **55** to which the controller **38** and the controller **40** are connected. The CPU **50**, the memory **51**, the storage unit **52**, the display unit **53**, the input unit **54**, and the external I/F **55** are connected to a bus **56**.

(24) The storage unit **52** is implemented with a hard disk drive (HDD), a solid state drive (SSD), a flash memory, and the like. A learning program **57**, a cell culture program **58**, and a learned model **59** are stored in the storage unit **52** as a storage medium. The CPU **50** reads out the learning program **57** and the cell culture program **58** from the storage unit **52**, extracts the learning program **57** and the cell culture program **58** to the memory **51**, and executes the extracted learning program **57** and cell culture program **58**.

(25) <Learning Phase>

(26) Next, a learning phase for obtaining the learned model **59** will be described. With reference to FIG. 4, a functional configuration of the information processing apparatus **12** according to the

present embodiment in the learning phase will be described. As shown in FIG. 4, the information processing apparatus **12** includes an acquisition unit **60**, a derivation unit **62**, and a generation unit **64**. The CPU **50** executes the learning program **57** to function as the acquisition unit **60**, the derivation unit **62**, and the generation unit **64**.

(27) The acquisition unit **60** acquires the transepithelial electrical resistance measured by the resistance measurement unit **34** for each cell culture channel **16**, via the controller **38**. Further, the acquisition unit **60** acquires image data showing the image of the cultured cells imaged by the imaging unit **36** for each cell culture channel **16**, via the controller **38**.

(28) In addition, the acquisition unit **60** acquires the evaluation value for the state of the cells that are cultured in the fluidic device **24** of each cell culture channel **16**, the culture environment, and information indicating the disposition position of the fluidic device **24** in the incubator **14** (hereinafter, referred to as “a disposition position information”).

(29) Specifically, the acquisition unit **60** acquires the flow rate of the liquid flowing through the fluidic device **24** per unit time, which is measured by the flow rate measurement unit **28**, as the culture environment of the cultured cells, via the controller **38**. Further, the acquisition unit **60** acquires the temperature of the liquid flowing through the fluidic device **24**, the environment temperature around the fluidic device **24**, the environment humidity around the fluidic device **24**, the carbon dioxide concentration around the fluidic device **24**, the nitrogen concentration around the fluidic device **24**, and the oxygen concentration around the fluidic device **24**, which are measured by the environment measurement unit **42**, as the culture environment of the cultured cells, via the controller **40**.

(30) In addition, the acquisition unit **60** acquires the evaluation value derived by the derivation unit **62**, which will be described later, as the evaluation value for the state of the cells. Further, the acquisition unit **60** acquires the disposition position information, for example, from the storage unit **52**. The disposition position information of the fluidic device **24** will be described in detail with reference to FIG. 5. For ease of understanding, FIG. 5 shows an example in which the cell culture system **10** is provided with four cell culture channels **16**.

(31) As shown in FIG. 5, in the present embodiment, as the disposition position information of the fluidic device **24**, information indicating the number from the top and information indicating the number from the left in a case where each cell culture channel **16** is viewed from the front of the incubator **14** are assigned to each cell culture channel **16**. For example, “1-1” is assigned as the disposition position information to the cell culture channel **16** which is the first from the top and the first from the left in FIG. 5. In the present embodiment, the disposition position information is associated with the identification information of the cell culture channel **16** and stored in the storage unit **52**. The identification information of the cell culture channel **16** is given, for example, in a case where the controller **38** outputs the transepithelial electrical resistance, the image data, and the flow rate acquired for each cell culture channel **16** to the information processing apparatus **12**.

(32) Accordingly, in a case where the acquisition unit **60** acquires each information regarding the flow rate of the liquid flowing through the fluidic device **24**, the transepithelial electrical resistance, and the image data, via the controller **38**, the acquisition unit **60** can specify which cell culture channel **16** each information relates to. Furthermore, the acquisition unit **60** can also specify which cell culture channel **16** disposed at which position in the incubator **14** each information relates to. The acquisition unit **60** periodically acquires various kinds of information described above.

(33) The derivation unit **62** derives the cell directions for the cultured cells of each cell culture channel **16** by using the image shown by the image data of each frame acquired by the acquisition unit **60**. The derivation unit **62** can derive the cell direction from the image of each frame by using the directionality of the cells of which the long-axes or short-axes converge into one direction, with respect to the flowing liquid.

(34) Further, the derivation unit **62** derives the cell migration distance from the image of each

frame and divides the migration distance by the time between frames to derive the cell migration speed. The cell direction and the cell migration speed are examples of information that is derived from the images obtained by imaging cells. Further, the cell direction, the cell migration speed, and the transepithelial electrical resistance are examples of information representing the state of the cells.

(35) The derivation unit **62** derives an evaluation value for the state of the cells, for the cultured cells of each cell culture channel **16**, on the basis of the transepithelial electrical resistance, the cell direction, and the cell migration speed. In the present embodiment, the derivation unit **62** derives a higher evaluation value, for example, as the numerical value, which is set to 10 levels from 1 to 10, increases, as the evaluation value for the state of the cells. The derivation unit **62** may derive a higher evaluation value, for example, as the convergence degree of the cell directions increases. Alternatively, the derivation unit **62** may derive a higher evaluation value, for example, as the cell migration speed increases. Alternatively, the derivation unit **62** may derive a higher evaluation value, for example, as the magnitude of transepithelial electrical resistance increases. In addition, the derivation unit **62** may use the shear stress quantification of the cells to derive the evaluation value for the state of the cells.

(36) For example, the user adjusts the culture environment for each cell culture channel **16** while checking the evaluation value for the state of the cultured cells. With this adjustment, for example, the evaluation values for the states of the cultured cells of the four cell culture channels **16** change. The generation unit **64** according to the present embodiment uses, as training data, the evaluation values before adjustment of the culture environment, the culture environment after the adjustment, and the disposition position information, in a case where, due to the adjustment, each evaluation value becomes a first threshold value or more and the absolute value of the difference in evaluation value between the cell culture channels becomes a second threshold value or less, among the evaluation values for the states of the cells, the culture environment, and the disposition position information. This is because it is preferable that the cells of which the evaluation value is a certain value or more and the variation in state is small can be obtained in each cell culture channel **16**.

(37) In a case where a first threshold value is set to 8 and a second threshold value is set to 0, the culture environment after adjustment of the culture environment, the disposition position information, and the evaluation values before the adjustment, for example, when due to the adjustment, the evaluation values for the states of the cultured cells of the four cell culture channels **16** are from 3, 4, 2, 5 to 8, 8, 8, 8, respectively, are used as the training data.

(38) The user repeatedly cultures the cells in the cell culture system **10**, so that the generation unit **64** can obtain a large number of training data. As shown in FIG. 6 as an example, the generation unit **64** generates the learned model **59** in which the evaluation value and the disposition position information for each cell culture channel **16** are received as an input and the culture environment is output, through the machine learning using the training data obtained as described above. The culture environment that is output by the learned model **59** includes the flow rate of the liquid flowing through the fluidic device **24** per unit time, the temperature of the liquid flowing through the fluidic device **24**, the environment temperature around the fluidic device **24**, the environment humidity around the fluidic device **24**, the carbon dioxide concentration around the fluidic device **24**, the nitrogen concentration around the fluidic device **24**, and the oxygen concentration around the fluidic device **24**. For example, the generation unit **64** applies a deep neural network as the learned model **59** to generate the learned model **59** by using the backpropagation method. The generation unit **64** may use a method other than the neural network, such as linear regression and a decision tree, as the machine learning method.

(39) Next, with reference to FIG. 7, the action of the information processing apparatus **12** according to the present embodiment in the learning phase will be described. The CPU **50** executes the learning program **57**, whereby learning processing shown in FIG. 7 is executed. The learning processing shown in FIG. 7 is executed, for example, in a case where the user inputs a start

instruction via the input unit **54**.

(40) In Step **S10** of FIG. 7, the acquisition unit **60** collects training data. As described above, the acquisition unit **60** periodically acquires the evaluation value for the state of the cultured cells, the culture environment, and the disposition position information, for each cell culture channel **16**. The acquisition unit **60** collects, as training data, the evaluation values before adjustment of the culture environment, the culture environment after the adjustment, and the disposition position information, in a case where, due to the adjustment, each evaluation value becomes a first threshold value or more and the absolute value of the difference in evaluation value between the cell culture channels becomes a second threshold value or less, among the plurality of evaluation values, the culture environment, and the disposition position information which are acquired.

(41) In Step **S12**, as described above, the generation unit **64** generates the learned model **59** in which the evaluation value and the disposition position information for each cell culture channel **16** are received as an input and the culture environment is output, through the machine learning using the training data collected by processing of Step **S10**. When the processing of Step **S12** ends, the learning processing ends.

(42) <Operation Phase>

(43) Next, an operation phase using the learned model **59** will be described. The operation phase is a phase in which cells are cultured in each cell culture channel **16**. With reference to FIG. 8, a functional configuration of the information processing apparatus **12** according to the present embodiment in the operation phase will be described. As shown in FIG. 8, the information processing apparatus **12** includes an acquisition unit **70**, a derivation unit **72**, and an integrated controller **74**. The CPU **50** executes the cell culture program **58** to function as the acquisition unit **70**, the derivation unit **72**, and the integrated controller **74**.

(44) As in the acquisition unit **60**, the acquisition unit **70** acquires transepithelial electrical resistance and image data for each cell culture channel **16**. Further, the acquisition unit **70** acquires the evaluation value for the state of the cells that are cultured in the fluidic device **24** of each cell culture channel **16** and the disposition position information of the fluidic device **24**, as in the acquisition unit **60**.

(45) As in the derivation unit **62**, the derivation unit **72** derives the evaluation value for the state of the cells for the cultured cells of each cell culture channel **16**.

(46) The integrated controller **74** adjusts the culture environment such that the states of the cultured cells of the cell culture channels **16** are the same, on the basis of the state of the cultured cells and the culture environment of each cell culture channel **16**. The integrated controller **74** according to the present embodiment adjusts the culture environment using the learned model **59** described above.

(47) Specifically, the disposition position information and the evaluation value for the state of the cells for each cell culture channel **16**, which are acquired by the acquisition unit **70**, are received to the learned model **59**, as an input, by the integrated controller **74**. Further, the integrated controller **74** acquires the culture environment output by the learned model **59** in response to the input.

(48) The integrated controller **74** performs control by outputting the culture environment that can be controlled for each cell culture channel **16** (in the present embodiment, the flow rate), among the acquired the culture environment, to the controller **38**. The controller **38** controls the pump **22** such that the flow rate of the liquid flowing through the fluidic device **24** becomes the flow rate received as an input from the integrated controller **74**.

(49) The integrated controller **74** performs control by outputting the culture environment that can be controlled for each incubator **14** (in the present embodiment, the culture environment that can be adjusted by the environment adjustment unit **44**), among the acquired the culture environment, to the controller **40**. The controller **40** controls the environment adjustment unit **44** such that the culture environment that is measured by the environment measurement unit **42** becomes the environment received as an input from the integrated controller **74**. That is, the integrated controller

74 adjusts the culture environment such that the absolute value of the difference between the evaluation values for the states of the cells that are cultured in the cell culture channels **16** is the second threshold value or less.

(50) Next, with reference to FIG. **9**, the action of the information processing apparatus **12** according to the present embodiment in the operation phase will be described. The CPU **50** executes the cell culture program **58**, whereby cell culture processing shown in FIG. **9** is executed. The cell culture processing shown in FIG. **9** is executed, for example, in a case where the user inputs a start instruction via the input unit **54**.

(51) In Step **S20** of FIG. **9**, as described above, the acquisition unit **70** acquires the evaluation value for the state of the cultured cells and the disposition position information for each cell culture channel **16**. In Step **S22**, as described above, the integrated controller **74** acquires the culture environment output by the learned model **59** in response to the input of the evaluation value and the disposition position information received to the learned model **59**, which are acquired by the processing of Step **S20**.

(52) In Step **S24**, as described above, the integrated controller **74** performs control by outputting the culture environment that can be controlled for each cell culture channel **16**, among the culture environment acquired by the processing of Step **S22**, to the controller **38**. Further, the integrated controller **74** performs control by outputting the culture environment that can be controlled for each incubator **14**, among the culture environment acquired by the processing of Step **S22**, to the controller **40**. With the processing of Step **S24**, the culture environment of the cells cultured in each cell culture channel **16** is adjusted. When the processing of Step **S24** ends, the cell culture processing ends. The above processing from Steps **S20** to **S24** is repeated at a predetermined time interval, for example, until the evaluation value for the state of the cultured cells of each cell culture channel **16** is the first threshold value or more and the absolute value of the difference in evaluation value between the cell culture channels is the second threshold value or less. The cells of which the evaluation value is a certain value or more and the variation in state is small, which are obtained by repeating the processing, are used for applications such as drug discovery.

(53) As described above, according to the present embodiment, the culture environment of each of the plurality of fluidic devices **24** can be adjusted in an integrated manner.

(54) In the above-described embodiment, the culture environment that can be adjusted for each incubator **14** by the environment adjustment unit **44** may be individually adjusted for each cell culture channel **16**. In this case, an aspect in which the environment measurement unit **42** and the environment adjustment unit **44** are provided in each cell culture channel **16** and controlled by the controller **38** of each cell culture channel **16** is exemplified. Further, in the example of the aspect, the culture environment that is output by the learned model **59** is also output corresponding to each cell culture channel **16**. In the example of the aspect, the generation unit **64** may generate the learned model **59** for each cell culture channel **16**. Further, in the example of the aspect, the culture environment of each cell culture channel **16** can be individually adjusted by the integrated controller **74**.

(55) Further, in the operation phase of the above embodiment, the integrated controller **74** of the information processing apparatus **12** may perform control to make the display unit **53** display the evaluation value for the state of the cultured cells of each cell culture channel **16**. In this case, an aspect in which the user switches to a manual operation to adjust the culture environment in a case where the evaluation value for the state of the cells displayed by the display unit **53** does not satisfy the desired condition, is exemplified.

(56) Further, in the above-described embodiment, for example, as a hardware structure of a processing unit that executes various processing such as processing performed by the acquisition units **60** and **70**, the derivation units **62** and **72**, the generation unit **64**, and the integrated controller **74**, the following various processors may be used. The various processors include, for example, a programmable logic device (PLD), such as an FPGA, which is a processor having a changeable

circuit configuration after manufacture and a dedicated electrical circuit, such as an application specific integrated circuit (ASIC), which is a processor having a dedicated circuit configuration designed to perform specific processing, in addition to the CPU which is a general-purpose processor that executes software (program) to function as various processing units as described above.

(57) One processing unit may be formed by one of the various processors or may be formed by a combination of two or more processors of the same type or different types (for example, a combination of a plurality of FPGAs and a combination of a CPU and an FPGA). Alternatively, the plurality of processing units may be formed by one processor.

(58) A first example of the configuration in which the plurality of processing units are formed by one processor is an aspect in which one or more CPUs and software are combined to form one processor and the processor functions as a plurality of processing units. A representative example of the aspect is a computer such as a client and server. A second example of the configuration is an aspect in which a processor that implements all of the functions of a system including the plurality of processing units with one integrated circuit (IC) chip is used. A representative example of the aspect is a system-on-chip (SoC). As described above, as the hardware structure of various processing units, one or more of the various processors are used.

(59) Furthermore, an electrical circuit (circuitry) obtained by combining circuit elements, such as semiconductor elements, may be used as the hardware structure of these various processors.

(60) In the above-described embodiment, the aspect in which the learning program **57** and the cell culture program **58** are stored (installed) in the storage unit **52** in advance has been described, but the present disclosure is not limited thereto. The learning program **57** and the cell culture program **58** may be recorded on a recording medium, such as a compact disc read only memory (CD-ROM), a digital versatile disc read only memory (DVD-ROM), or a universal serial bus (USB) memory, and then provided. Alternatively, the learning program **57** and the cell culture program **58** may be downloaded from an external apparatus via the network.

(61) In the cell culture system according to the present disclosure, the integrated controller may individually adjust the culture environment of the cells which are cultured in the plurality of cell culture channels, for each cell culture channel.

(62) Further, in the cell culture system according to the present disclosure, the integrated controller may adjust the culture environment by using a learned model in which disposition position information representing a disposition position of the cell culture channel and the evaluation value for the state of the cells, which is determined from the state of the cells, are received as an input and the culture environment is output.

(63) In addition, in the cell culture system according to the present disclosure, the state of the cells may include information that is derived from an image obtained by imaging the cells and transepithelial electrical resistance of the cells.

(64) Further, in the cell culture system according to the present disclosure, the culture environment may include a flow rate of the liquid, a temperature of the liquid, an environment temperature around the fluidic device, an environment humidity around the fluidic device, a carbon dioxide concentration around the fluidic device, a nitrogen concentration around the fluidic device, and an oxygen concentration around the fluidic device.

(65) According to the present disclosure, the culture environment of each of the plurality of fluidic devices can be adjusted in an integrated manner.

Claims

1. A cell culture system comprising: a plurality of cell culture channels each of which includes a fluidic device in which cells are cultured, a liquid feeding unit which makes a liquid flow into the fluidic device, a first sensor which detects a state of the cells that are cultured in the fluidic device,

and a first processor which controls the liquid feeding unit and the first sensor; a second sensor that measures culture environment of the cells which are cultured in the plurality of cell culture channels; and a second processor, wherein the second processor is configured to: compute evaluation values corresponding to each of the plurality of cell culture channels for the states of the cells that are cultured in each of the plurality of cell culture channels, on the basis of the state of the cells detected by the first sensor and the culture environment of the cells measured by the second sensor; compute an absolute value of a difference in the evaluation values between different ones of the plurality of cell culture channels; and adjusts the culture environment such that the computed absolute value is a threshold value or less.

2. The cell culture system according to claim 1, wherein the second processor individually adjusts the culture environment of the cells which are cultured in the plurality of cell culture channels, for each cell culture channel.

3. The cell culture system according to claim 1, wherein the second processor adjusts the culture environment by using a learned model in which disposition position information representing a disposition position of the cell culture channel and the evaluation value for the state of the cells, which is determined from the state of the cells, are received as an input and the culture environment is output.

4. The cell culture system according to claim 2, wherein the second processor adjusts the culture environment by using a learned model in which disposition position information representing a disposition position of the cell culture channel and the evaluation value for the state of the cells, which is determined from the state of the cells, are received as an input and the culture environment is output.

5. The cell culture system according to claim 1, wherein the state of the cells includes information that is derived from an image obtained by imaging the cells and transepithelial electrical resistance of the cells.

6. The cell culture system according to claim 2, wherein the state of the cells includes information that is derived from an image obtained by imaging the cells and transepithelial electrical resistance of the cells.

7. The cell culture system according to claim 3, wherein the state of the cells includes information that is derived from an image obtained by imaging the cells and transepithelial electrical resistance of the cells.

8. The cell culture system according to claim 1, wherein the culture environment includes a flow rate of the liquid, a temperature of the liquid, an environment temperature around the fluidic device, an environment humidity around the fluidic device, a carbon dioxide concentration around the fluidic device, a nitrogen concentration around the fluidic device, and an oxygen concentration around the fluidic device.

9. The cell culture system according to claim 2, wherein the culture environment includes a flow rate of the liquid, a temperature of the liquid, an environment temperature around the fluidic device, an environment humidity around the fluidic device, a carbon dioxide concentration around the fluidic device, a nitrogen concentration around the fluidic device, and an oxygen concentration around the fluidic device.

10. The cell culture system according to claim 3, wherein the culture environment includes a flow rate of the liquid, a temperature of the liquid, an environment temperature around the fluidic device, an environment humidity around the fluidic device, a carbon dioxide concentration around the fluidic device, a nitrogen concentration around the fluidic device, and an oxygen concentration around the fluidic device.

11. The cell culture system according to claim 5, wherein the culture environment includes a flow rate of the liquid, a temperature of the liquid, an environment temperature around the fluidic device, an environment humidity around the fluidic device, a carbon dioxide concentration around the fluidic device, a nitrogen concentration around the fluidic device, and an oxygen concentration

around the fluidic device.

12. A cell culture method using a cell culture system that includes a plurality of cell culture channels each of which includes a fluidic device in which cells are cultured, a liquid feeding unit which makes a liquid flow into the fluidic device, a first sensor which detects a state of the cells that are cultured in the fluidic device, a first processor which controls the liquid feeding unit and the first sensor, a second sensor that measures culture environment of the cells which are cultured in the plurality of cell culture channels, and a second processor, the cell culture method comprising: by the second processor: computing evaluation values corresponding to each of the plurality of cell culture channels for the states of the cells that are cultured in each of the plurality of cell culture channels, on the basis of the state of the cells detected by the first sensor and the culture environment of the cells measured by the second sensor; computing an absolute value of a difference in the evaluation values between different ones of the plurality of cell culture channels; and adjusting the culture environment such that the computed absolute value is a threshold value or less.
